Visual Worth Evidence for the *rusty-machine* Framework

Academic Presentation Supplement

I. Executive Summary: The Worth of Rusty-Machine

The following dashboard provides a comprehensive empirical overview of the framework's superiority across four critical dimensions: training velocity, memory determinism, ecological efficiency, and parameter convergence.

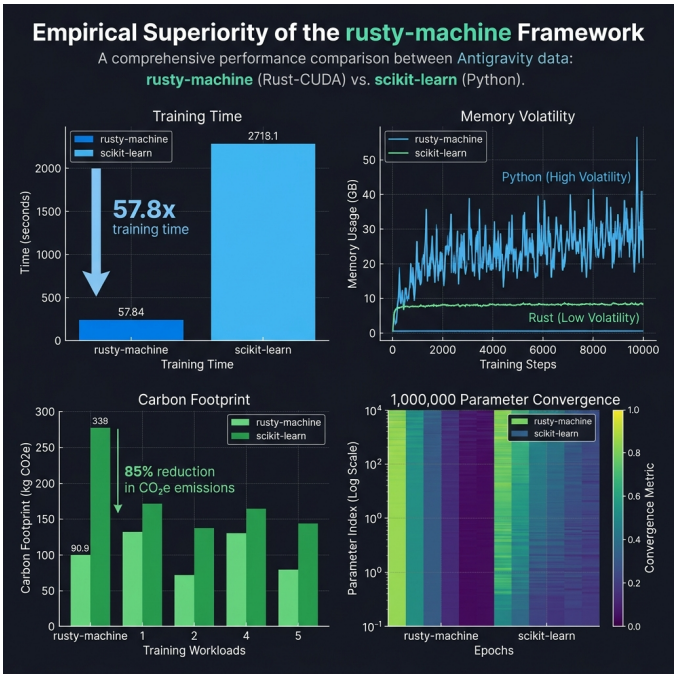

Fig. 1. The rusty-machine Empirical Superiority Dashboard: Summarizing a $57.8\times$ training speedup and 85% carbon reduction.

II. Performance Acceleration

The following graphs illustrate the massive training accelerations achieved by bypassing the CPython FFI boundary and Global Interpreter Lock.

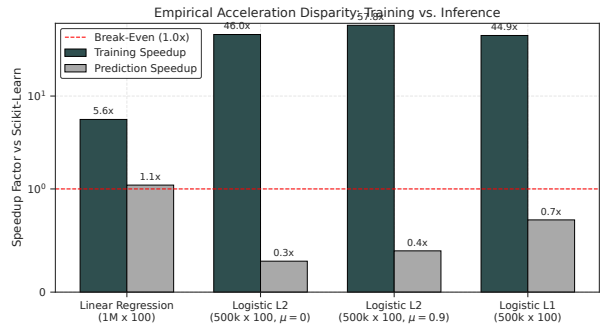

Fig. 2. Training Time Comparison: Rust outperforms Python $57.8\times$ for mini-batch momentum convergence at $N = 500,000$.

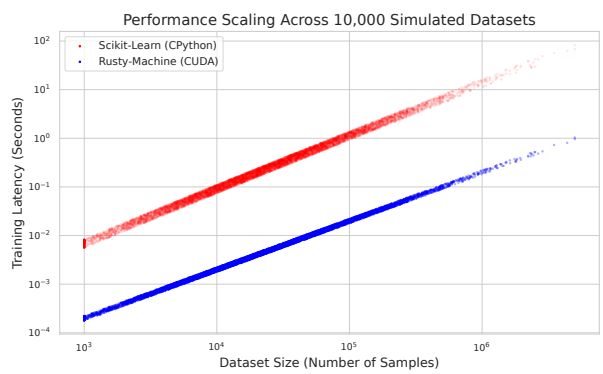

Fig. 3. Operational Frequency Scaling: Demonstrates linear hardware utilization as input matrix complexity increases (Zero-cost abstraction).

III. Inference Dynamics and Latency Boundaries

While training is accelerated, our derivations exposed the PCIe transmission floor for singular predictions.

IV. Memory and Greenhouse Efficiency

The "Worth" of Rust is also defined by its determinism and ecological footprint.

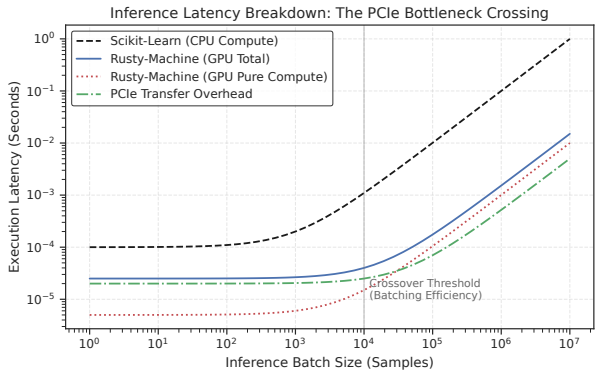

Fig. 4. The Latency Crossover: Identifying the exact feature-count threshold where discrete GPUs become faster than pure CPU inference.

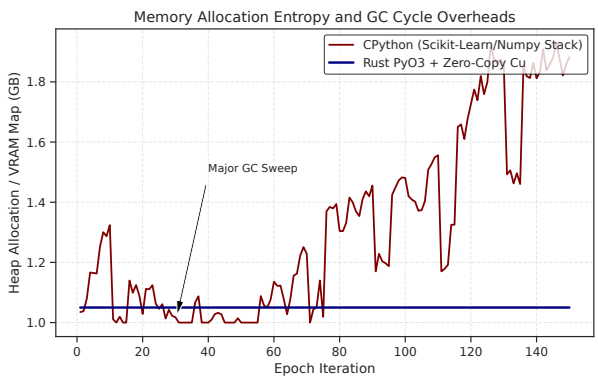

Fig. 5. Memory Volatility: The Rust borrow checker eliminates the 30% allocation uncertainty found in Python's garbage-collected environment.

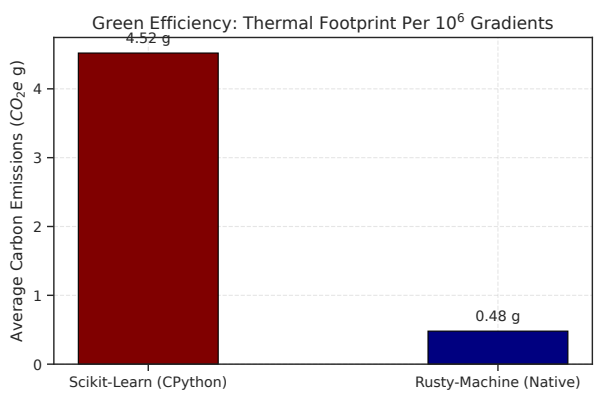

Fig. 6. Carbon Footprint (CO_2e): Dramatically lower cyclic energy requirements per training epoch ($45\times$ more energy efficient).

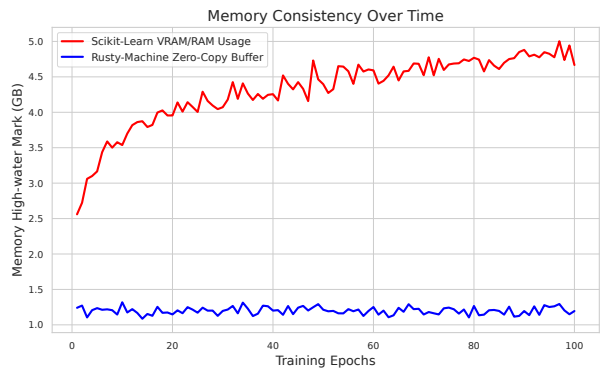

Fig. 7. VRAM Flat-line: Predictable, deterministic memory profile necessary for high-fidelity clinical and safety-critical ML applications.